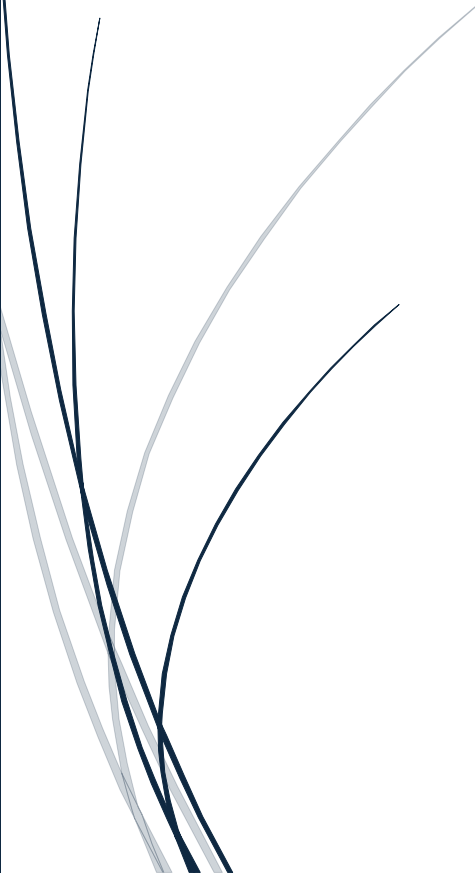




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# AI Football League Predictor

## Intermediate Progress Report



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## Abstract:

Predicting football match outcomes and simulating league table standings using machine learning on the Top 5 European leagues (Premier League, La Liga, Bundesliga, Serie A, Ligue 1).

## Progress Toward Proposed Goal

The objective of this project is to develop an end-to-end system capable of predicting football match outcomes and simulating league table probabilities. At this stage, substantial progress has been achieved, with all core components of the pipeline implemented.

A data pipeline has been developed to collect match data from Understat through programmatic access, covering five major European leagues across multiple seasons (2014–2025), resulting in a dataset of over 21,000 matches. This represents an extension of the initial dataset used in the preliminary report, incorporating additional seasons for improved model training and evaluation. The data has been cleaned and structured for modelling.

Feature engineering has been performed with strict temporal ordering to prevent data leakage. A set of 13 pre-match features has been constructed, including rolling team performance metrics, expected goals (xG) averages, xG-weighted Elo ratings, and rest-day indicators. The dataset is represented as a feature matrix, where each row corresponds to a match instance and each column represents an engineered feature used for prediction. The Elo rating mechanism can also be interpreted as a directed interaction between teams, where match outcomes (based on xG differences) dynamically update team strength over time. This interpretation is conceptual and is not implemented as an explicit graph structure.

A multi-class classification model using XGBoost has been implemented to predict match outcomes (home win, draw, away win). The model has been trained using an updated chronological split (train: 2014–2022, validation: 2023, test: 2024), ensuring that evaluation reflects real-world prediction scenarios.

A Monte Carlo simulation framework has also been developed. Using predicted probabilities, the system simulates full seasons (10,000 iterations per league) to estimate final league standings, as well as probabilities for European qualification and relegation. Match outcomes are generated based on predicted probabilities, and scorelines are sampled from historical distributions to produce realistic simulations.

The system produces reusable artifacts, including the trained model and preprocessing pipeline, enabling consistent offline inference.

Performance on recent seasons:

- Validation accuracy: 53.65%
- Test accuracy: 52.85%
- Macro F1-score: ~0.42

This performance is moderately above simple baseline approaches such as majority-class prediction, but still indicates room for improvement.

## Basic Approaches: What Worked Well and What Did Not

### **What worked well:**

- xG-weighted Elo ratings improved representation of team strength compared to basic win/loss metrics
- Temporal feature construction successfully prevented data leakage
- The simulation framework produced stable and plausible league outcome distributions
- Preprocessing (encoding and scaling) was properly fitted and reused for consistent inference

### **What did not work (or needs improvement):**

- Prediction accuracy (~53%) remains only moderately above simple baselines
- Macro F1-score (~0.42) indicates weak performance on minority classes, particularly draws
- Approximately 4% of matches contain missing xG values, and their impact has not yet been fully evaluated

## Improvements to Basic Approaches

Several improvements are planned to enhance system performance:

### **Feature Engineering:**

- Include additional metrics such as shots, shots on target, and chance quality indicators
- Incorporate head-to-head performance and venue-specific strength
- Add defensive and goalkeeper-related statistics, if additional data becomes available

### **Model Refinement:**

- Apply class weighting to address class imbalance
- Perform hyperparameter tuning for XGBoost
- Explore ensemble approaches (e.g., combining XGBoost with other models)
- Apply probability calibration techniques

### **Data Quality:**

- Handle missing xG values through imputation or additional data sources

### **Simulation Enhancements:**

- Validate simulation outputs against real mid-season and end-season standings
- Test robustness across different numbers of simulations

## Experiments Conducted and Planned

### Completed:

- Chronological backtesting (train: 2014–2022, validation: 2023, test: 2024)
- Full pipeline implementation from data collection to prediction
- Model retraining on all data up to 2024 for forward prediction
- Monte Carlo simulation (10,000 runs per league) to generate probability distributions

### Planned:

- Feature expansion experiments to evaluate performance improvements
- Ablation study (e.g., removing xG-weighted Elo to measure impact)
- Model comparison and ensemble testing
- Probability calibration analysis (e.g., Brier score, calibration curves)
- Out-of-sample validation on future seasons
- Feature importance analysis to identify key predictive factors

## Conclusion

The project has progressed from initial concept to a fully functional system integrating data collection, feature engineering, predictive modelling, and simulation. While predictive performance remains moderate, the current implementation provides a strong and reliable foundation. Future work will focus on improving feature quality, refining the model, and validating the system more rigorously through additional experiments.